

Pre- and Post-use Sanitization Methods

Millistak+® Pod filters are currently used at various points in the bioprocess template including primary and secondary clarification, post affinity chromatography haze removal and as a prefilter for virus filtration. While other studies conducted with Millistak+® media demonstrated very low levels of initial bioburden as well as the effectiveness of the recommended flushing procedures, a need has been identified for pre-use sanitization guidelines. Several studies were conducted to evaluate the feasibility to implement sanitization methods commonly used in the bioprocess industry. These methods include the following:

- Hot water at 80 °C
- 0.5 N and 1.0 N Sodium Hydroxide
- Autoclaving of Pod filters installed into a holder (please note: modified hardware is required for this option)

Sodium Hydroxide (NaOH) Sanitization

This study aims at understanding and confirming the impact of 0.5 N and 0.1 N NaOH sanitization at room temperature. The effect of NaOH sanitization on Millistak+® Pod filters, as well as performance of the clarification train, was studied. The following parameters were used as study metrics:

1. Depth filters capacity
2. Sterile filter capacity
3. TOC levels in flush samples after sanitization
4. Conductivity levels in flush samples after sanitization

Materials and Test Method

Control and sanitized filters were used from the same lot for each sanitizing condition (Table 11). The mixture of whey solution and CHO cells was prepared in the lab to mimic the typical turbidity observed in cell culture harvest. Different concentrations of ingredients were used to make loading samples for Millistak+® A1HC and X0HC filters based on targeted loading of 300 L/m². Feed composition was changed for 0.5 N NaOH sanitization study based on the plugging observed in 0.1 N sanitization study. The formulations are shown in Table 12.

Table 11. Material Information

Test Material	Lot No.	Catalogue No.
Millistak+® Pod A1HC 0.027 m ²	CP9NN79372	MA1HC027H1
Millistak+® Pod A1HC 0.027 m ²	CP9PN79809	MA1HC027H1
Millistak+® Pod X0HC 0.027 m ²	CP9NN79374	MX0HC027H1
Millistak+® Pod X0HC 0.027 m ²	CP9CN73418	MX0HC027H1
Durapore® CVGL 13.8 cm ²	R6AN42605	GVWP04700
Sigma-Aldrich whey	115K0037	W1500-2
Sigma-Aldrich PBS	059K8215	P4417-100TAB

Table 12. Feed Formulations

NaOH Concentration	A1HC	X0HC
0.1 N NaOH	5 g/L whey	5 g/L whey
	5 g/L PBS	5 g/L PBS
	20 mL/L CHO cells	20 mL/L CHO cells
Turbidity	81.5 NTU	44.5 NTU
0.5 N NaOH	10 g/L whey	2.5 g/L whey
	5 g/L PBS	5 g/L PBS
	20 mL/L CHO cells	20 mL/L CHO cells
Turbidity	170 NTU	50.9 NTU

Test Method

1. Wet media at 600 LMH (270 mL/min for 270 cm²) for 100 L/m² (2.7 L for 270 cm²)
2. Perform sanitization at 300 LMH (135 mL/min for 270 cm²). Recirculate for 1 hour at room temperature.
3. Flush filter with room temperature water at 300–600 LMH until a filtrate conductivity of <10 µS/cm is achieved. Take samples every 100 L/m² for TOC and conductivity analysis.
4. Perform $P_{\max}^{\text{sm}}$ capacity experiment at 150 LMH (67.5 mL/min for 270 cm²) using whey and CHO cell solution to a final pressure of 20 psid (1.38 bar).
5. Perform $V_{\max}^{\text{sm}}$ capacity experiment on the Millistak+® Pod filtrate using Durapore® 0.22 µm membrane (47 mm disk holder).

Filters were run at a constant flux of 150 LMH using a peristaltic pump. Filters were evaluated with lab scale Pod filters (270 cm²) with pressure transducers upstream of each filter to monitor filter plugging. Pressure was monitored throughout the run using the EMD Millipore Data Acquisition system and filtrate turbidity was measured using a Hach turbidity meter.

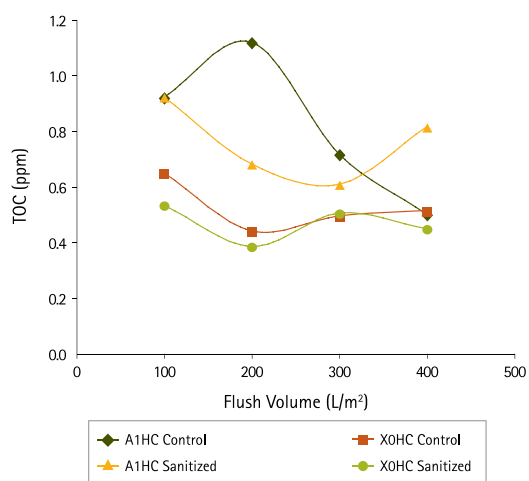


Figure 16.

TOC levels in the post 0.1 M NaOH sanitization water flush

Results and Discussion

TOC and Conductivity

Figures 16 and 17 show that after sanitization, the Total Organic Carbon (TOC) levels are similar for both control and sanitized filter flush samples. In both cases sanitized A1HC filter flush samples show slightly higher TOC. The TOC values of the water, which was used for flushing, was 0.506 ppm and 0.306 ppm in 0.1 N and 0.5 N sanitization, respectively.

Conductivity has historically been used to track the removal of extractables during the pre-use water flush. Since the sanitization solutions have a high conductivity, conductivity values seen in flush samples cannot be used to assess residual levels of extractables. Use of a buffer at the end of or for the entire pre-use flush should equilibrate the media to desired pH and conductivity before introduction of product.

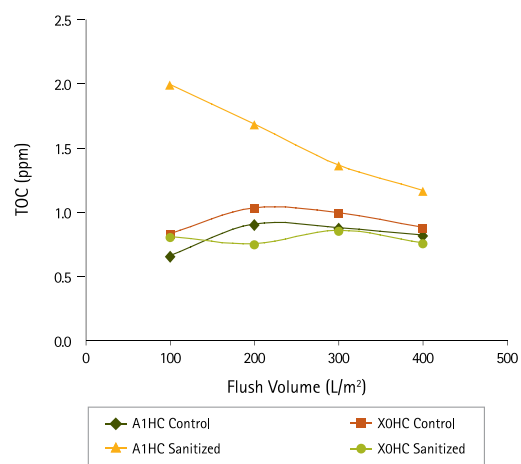


Figure 17.

TOC levels in the post 0.5 M NaOH sanitization water flush

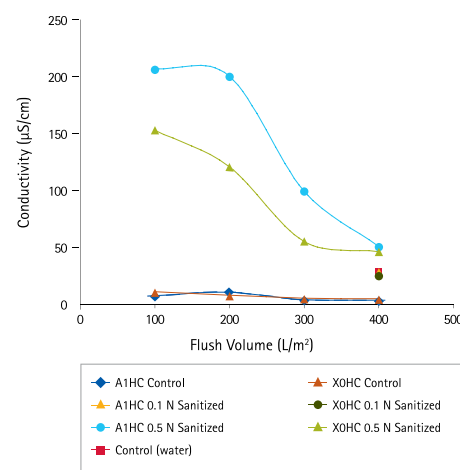


Figure 18.

Conductivity levels in 0.1 N and 0.5 N NaOH post sanitization flush

$P_{\max}^{\text{SM}}$ (Capacity vs. Resistance) Testing

Figures 19 and 20 show there was no major difference between the resistance vs. capacity plots of any pair of control and sanitized filters. As shown in Table 13, the resistance curves of paired Millistak+® Pod filters matched closely throughout the filtration trial. The variation in the capacities of each pair of Millistak+® Pod filters (sanitized and control) was $\leq 15\%$. The filtrates from all the Pod runs were non-fouling on a 0.22 μm sterilizing grade filter. Figure 21 shows that no significant difference was seen in the mean filtrate flux from feeds of sanitized or control Pods.

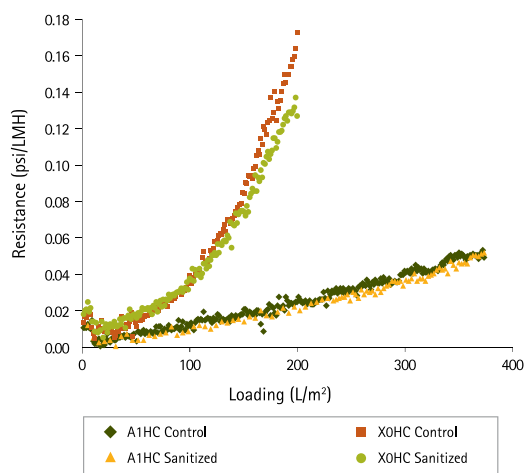


Figure 19.
Resistance vs. capacity post 0.1 N NaOH sanitization

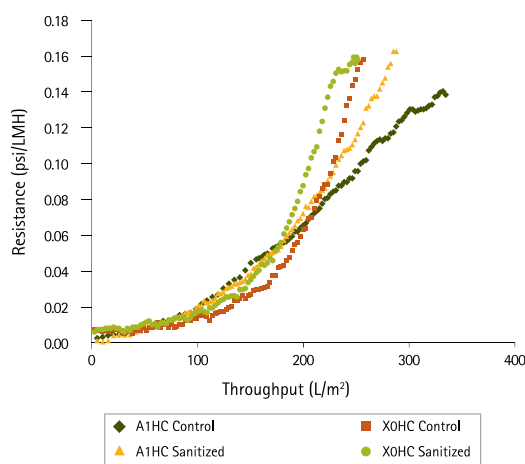


Figure 20.
Resistance vs. capacity post 0.5 N NaOH sanitization

Table 13. Pod Capacities at 20 psid (1.38 bar)

Millistak+® Filter	Capacity (L/m²) at 20 psid
A1HC control	373
A1HC 0.1 N NaOH sanitized	373
X0HC control	176
X0HC 0.1 N NaOH sanitized	190
A1HC control	306
A1HC 0.5 N NaOH sanitized	259
X0HC control	241
X0HC 0.5 N NaOH sanitized	220

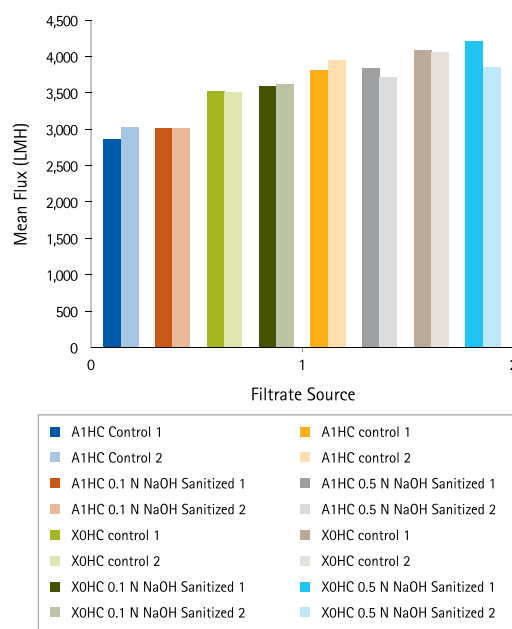


Figure 21.
Mean filtrate flux for 0.22 μm sterile filtrate on Millistak+® Pod filtrates

Conclusions and Recommendations

From the presented $P_{\max}^{\text{SM}}$ and $V_{\max}^{\text{SM}}$ data, no significant effect of NaOH sanitization (0.1 N and 0.5 N) on Millistak+® A1HC and X0HC Pod filters was observed. Performance was similar for both Millistak+® Pod and Durapore® 0.22 μm sterile filters. As a result, we can conclude that 0.1 N NaOH at room temperature for 1 hour and 0.5 N NaOH at room temperature for 1 hour did not negatively impact the performance of the Millistak+® Pod filters based on the model feed stream utilized in this study. In this study, the tested Millistak+® media was flushed with water prior to sanitization. For a typical installation, it should be possible to proceed directly with the sanitization step without the need for prior water flush.

Autoclaving Millistak+® Pod in a Pod Holder

Pod devices are rated for one autoclave cycle of 60 minutes at 123 °C; however, this claim is based on single devices being autoclaved alone (for example, in a Tyvek® bag) rather than on an autoclave procedure performed on devices installed in a holder. This study explores the feasibility to install devices in a holder, autoclave the entire assembly and then make an aseptic connection to a harvest system.

The purpose of this study is to:

- Determine suitable autoclave cycle parameters (cycle type, time and temperature) which can be used to sterilize Millistak+® Pod devices installed in a modified holder from which the hydraulics have been removed.
- Verify the feasibility of autoclaving Millistak+® Pod filters in a modified holder. "Feasibility" is demonstrated in two ways:
 1. Sterility can be achieved on a complete autoclaved Millistak+® Pod assembly.
 2. The autoclave cycle does not affect the integrity of the devices as determined by an air-water pressure hold test.

Results of this study should be taken as demonstrating compatibility of the Millistak+® Pod filters with such a process; however, the results do not constitute a complete validation of the process. As with any autoclave procedure, the burden of validation rests with the end user.



Figure 22.
Location of spore strips inside Millistak+® Pod filter

Experimental Design

The experimental procedure was divided into three subsections:

1. Modification of the Millistak+® Pod holder hardware to render it compatible with autoclaving.
2. Temperature mapping experiments designed to identify "cold spots" or areas that lag behind the chamber temperature in the autoclave.
3. Spore kill experiments designed to show that heat-resistant bacteria can be rendered sterile in an autoclave cycle.

Millistak+® A1HC Pod devices were used in all experiments because they contain the least permeable filtration media, providing the worst case for steam penetration.

Temperature mapping experiments were performed using a GE Kaye Instruments Validator® system. Millistak+® Pod devices were modified by drilling 0.313 inch diameter holes in four locations:

1. In the feed channel behind the blanked port located opposite the feed port
2. In the permeate channel behind the blanked port opposite the permeate port
3. Along the vertical centerline of the device at the lowest point, in the feed channel between plates one and two
4. Along the vertical centerline of the device at the lowest point, in the permeate channel between plates four and five

Thermocouples were installed into the holes and the holes were then sealed using rubber stoppers. The stoppers were secured using autoclave tape.



Figure 23.
Millistak+® Pod pilot holder

Table 14. Autoclave Cycle Parameters

Cycle Time	60 min
Cycle Temperature	123.1 °C
Pre-vacuum	80 °C
Post-vacuum	Three pre-vacuum cycles

Spore kill experiments were performed on specially manufactured devices which included spore strips located in four locations:

1. In the feed channel between plates five and six and behind the blanked port located opposite the feed port
2. In the permeate channel between plates five and six and behind the blanked port located opposite the permeate port
3. In the same corner as strip two, but between the 60DE and 75DE layers
4. In the same corner as strip two, but between the 75DE and RW01 layers

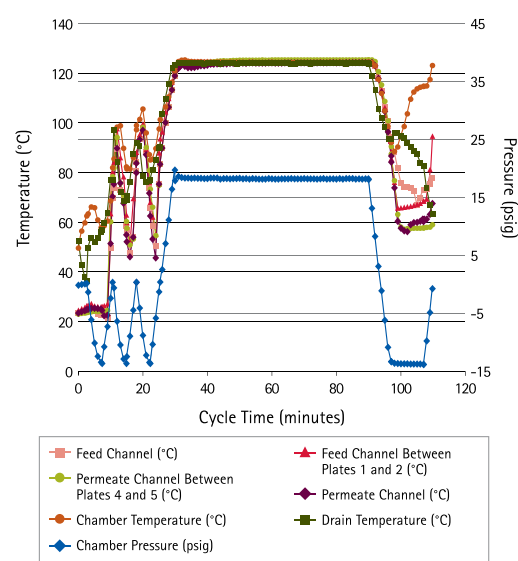


Figure 24.
Single 1.1 m² Millistak+® Pod device temperature mapping during autoclave cycle

Temperature Mapping Experiments

Figures 24 and 25 show the results of the temperature mapping experiments performed on 1.1 m² and 5.5 m² installations, respectively. As seen in the temperature profiles, no significant lag was observed at any location in the device, regardless of the installation size.

From this data, it was determined that a standard validation approach would be suitable for the spore kill experiments without providing any additional cycle time for steam penetration. (See "Spore Kill Experiments" on page 22.)

From the 5.5 m² installation, the three unmodified devices were removed from the holder, individually flushed with water and then tested for integrity using an air-water pressure hold test. Flushing was performed at 600 LMH with 100 L/m² of water and the pressure hold test was performed at approximately 3 to 3.5 psi (0.2 to 0.24 bar). Please note that flushing and testing were performed per the recommendations described in EMD Millipore Application Note, "Using Millistak+® HC Filters for Mammalian Cell Culture Clarification" (AN1100EN00). All devices passed the pressure hold test, indicating that the autoclave cycle did not damage the filter media.

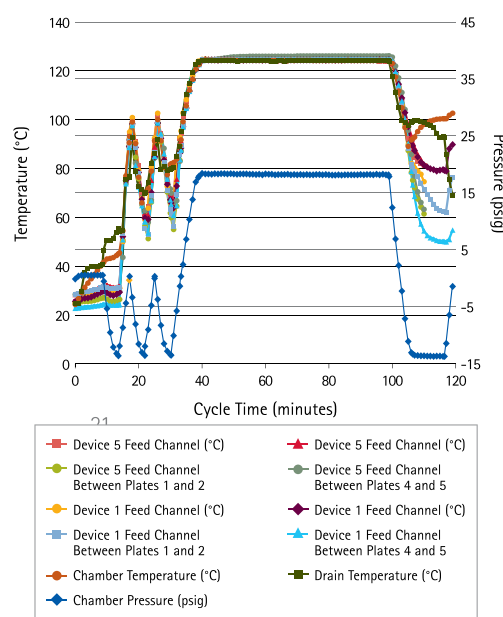


Figure 25.
Five 1.1 m² Millistak+® Pod devices temperature mapping during autoclave cycle